Business Problem Statement

A leading retail company aims to gain deeper insights into customer shopping behavior to improve sales performance, enhance customer satisfaction, and build long-term customer loyalty. Recently, the management team has observed noticeable shifts in purchasing patterns across different customer demographics, product categories, and sales channels (online vs. offline).

The company wants to understand the key factors that influence purchasing decisions and repeat purchases. These factors may include discounts, customer reviews, seasonal trends, and preferred payment methods.

The main objective of this project is to analyze the company's consumer behavior dataset to answer the following business question:

"How can the company use consumer shopping data to identify purchasing trends, improve customer engagement, and optimize marketing and product strategies?"

Deliverables

1. Data Preparation & Modeling (Python)
Clean, preprocess, and transform the raw dataset to make it suitable for analysis.

2. Data Analysis (SQL)
Structure the data in a relational format, simulate business transactions, and run SQL queries to uncover insights related to customer segments, loyalty patterns, and key purchase drivers.

3. Visualization & Insights (Power BI)
Develop an interactive dashboard that highlights important trends, customer behavior patterns, and business insights to support data-driven decision making.

4. Report and Presentation
Prepare a concise project report summarizing the analysis, key findings, and business recommendations. Create a presentation that clearly communicates insights and actionable strategies to stakeholders.

5. GitHub Repository
Maintain a well-organized GitHub repository containing all project assets, including Python scripts, SQL queries, datasets, and Power BI dashboard files.